AI Programming Assignment 2

Instructions

1. **Attempt all the questions. Show all your workings where applicable.**
2. **Work in the previously formed groups.**
3. **Submit a pdf version of your work at 2359Hrs on 04/11/2025.**

Questions (30 Marks)

a) Briefly explain any three (3) reasons why LISP is used for AI programming? (3 Marks)

b) Show the syntax of defining a function in LISP (3 Marks)

c) What are Macros in LISP and how do they work? (3 Marks)

d) Explain some four (4) commonly used data structures in Lisp? (6 Marks)

e) Explain what is CLOS in LISP? What are the goals of the common lisp object system? (4 Marks)

f) How does Lisp handle error conditions and exceptions? (3 Marks)

g) What are some common debugging techniques used in Lisp? (4 Marks)

h) How does Lisp handle input/output operations? (4 Marks)